

with the European Union's AI Act serve as guidelines for conducting safe experimental work. Researchers can perform device testing without unacceptable risks by using safety protocols and rollback mechanisms, and controlled testing environments that combine these frameworks.

The implementation of chaos engineering in IoT systems demands detailed planning, robust safety protocols and ethical supervision to succeed.

Towards a resilient IoT future

Open source communities are actively working to enhance IoT device dependability through their contributions of testing tools, platforms and research projects. An open chaos toolkit for IoT (e.g., software to simulate failures on smart sensors, wearables or industrial controllers) would enable basic resilience tests on devices. The system could operate through a structure that mirrors Linux operating system accessibility (common OS for IoT gateways and edge devices) and Kubernetes cloud system management (used to manage IoT services and applications at scale).

Multiple initiatives are currently trying to achieve this goal. The EdgeX Foundry (framework for connecting IoT devices from different vendors) and Eclipse IoT platforms (open source projects for managing IoT applications and devices) serve as frameworks that enable different devices to operate harmoniously together. Through GitHub, developers can share and test microcontrollers and embedded systems by running chaos simulation experiments. The built-in testing capabilities of Fledge (industrial IoT data platform for edge devices) and Zephyr RTOS (real-time operating system used in small IoT devices) platforms demonstrate successful methods to enhance edge device reliability. The development of IoT system reliability across device networks can improve through

collaborative development approaches used by the community.

The tools need benchmarks and standards support to operate correctly. The IoT device certification process evaluates how devices operate while also testing their ability to function through disruptions. The certification process for smart sensors (such as temperature or motion sensors) requires them to maintain data accuracy at 30% network packet loss rates. A wearable device needs to restore its full operational capabilities

within a five-second time frame after the system restarts. Users can verify device performance through such standardised benchmarks.

Teaching people about chaos engineering is just as important as developing tools. Most IoT developers focus on adding new features rather than preparing for failures. Open source initiatives, along with tutorials, workshops and hackathons, help train developers to run safe reliability tests and evaluate their devices' performance under stress. **END** 

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